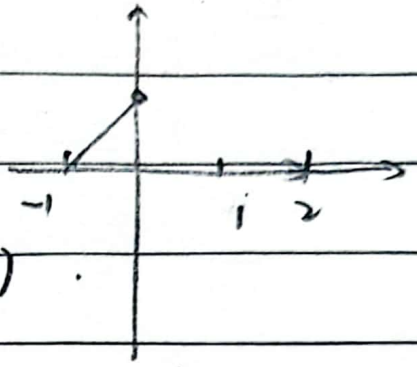
思考题四 6 7 10

习题四 4(1)(4)(5) 5(1) 10 12 13(2)(6)

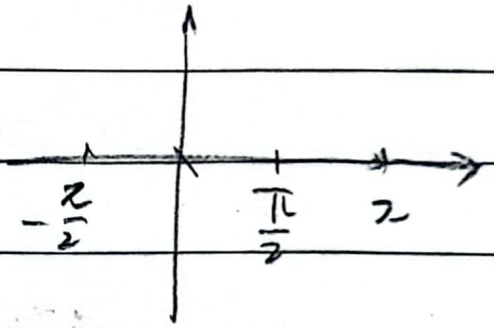
6、 $R=|z|$



7、 $\cos z=0 \Rightarrow e^{iz}=e^{-iz}=e^{i(\pi-z)}$

$z=\pi-z+2k\pi \Rightarrow z=\frac{\pi}{2}+k\pi$

$R=\frac{\pi}{2}$



10 $f(z)=\frac{1}{z(z^2-1)}$

$z=0$, $R_1<|z|<R_2$, $0\le R_1<R_2\le 1$, $\sum_{n=-\infty}^{+\infty} C_n z^n$

或 $R_1\ge 1$



$z=1$, $R_1<|z-1|<R_2$, $0\le R_1<R_2\le 1$, $\sum_{-\infty}^{+\infty} C_n (z-1)^n$

或 $1\le R_1<R_2\le 2$, 或 $R_1\ge 2$

$z=-1$, $R_1<|z+1|<R_2$, $0\le R_1<R_2\le 1$, $\sum_{-\infty}^{+\infty} C_n (z+1)^n$

或 $1\le R_1<R_2\le 2$ 或 $R_1\ge 2$

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$\sum x^n = \frac{1}{1-x}$ $\sum (-1)^n x^n = \frac{1}{1+x}$

$\sum n x^{n-1} = \frac{1}{(1-x)^2}$

$\left(\frac{1}{1-x}\right)' = \frac{1}{(1-x)^2} = \sum n x^{n-1}$

4. (1) $\frac{1}{z+1} = \frac{1}{2+z-1}$

$= \frac{1}{2} \frac{1}{1+\frac{z-1}{2}} = \frac{1}{2} \sum (-1)^n \frac{(z-1)^n}{2^n} =$

故 $\frac{1}{z+1} = \sum (-1)^n \frac{(z-1)^{n+1}}{2^{n+1}}$

(4) $\frac{1}{(z-2)^2} = \frac{1}{[1-(z-1)]^2} = \left[\frac{1}{1-(z-1)}\right]' = \left[\sum_{n=0}^{\infty} (z-1)^n\right]'$

$= \sum_{n=0}^{\infty} n (z-1)^{n-1}$ $(|z-1|<1)$.

(5) $\frac{1}{z^2-2z+10} = \frac{1}{(z-1)^2+9} = \frac{1}{9} \frac{1}{\left(\frac{z-1}{3}\right)^2+1} = \frac{1}{9} \sum (-1)^n \frac{(z-1)^{2n}}{3^{2n}}$

$\left|\left(\frac{z-1}{3}\right)^2\right| < 1 \Rightarrow |z-1| < 3$

5(1) $\frac{1}{(1+x)^2} = \sum_{n=0}^{\infty} n(-1)^n x^{n-1}$

故 $\frac{1}{(1+z^2)^2} = \sum_{n=0}^{\infty} n(-1)^n z^{2n-2}$ $|z^2|<1 \Rightarrow |z|<1$

10. $f(z) = e^{z^2} = 1 + z^2 + \frac{z^4}{2} + \cdots + \frac{z^{2n}}{n!} + \cdots$

故 $f^{(n)}(0) = \frac{(2n)!}{n!}$

12. (1) $f^{(n)}(z_0) = \frac{n!}{2\pi i}\int \frac{f(z)}{(z-z_0)^{n+1}} dz$

$f(\zeta) = e^{z\zeta}$, $\zeta_0 = 0$, 则 $z^n = \frac{n!}{2\pi i}\int \frac{e^{z\zeta}}{\zeta^{n+1}} d\zeta$

$\Rightarrow \left(\frac{z^n}{n!}\right)^2 = \frac{1}{2\pi i}\oint_{|\zeta|=1} \frac{z^n e^{z\zeta}}{n!\,\zeta^n} \frac{d\zeta}{\zeta}$

(2) $\sum_{n=0}^{\infty}\left(\frac{z^n}{n!}\right)^2 = \frac{1}{2\pi i}\oint_{|\zeta|=1} \frac{\sum \frac{z^n}{n!\zeta^n} e^{z\zeta}}{\zeta} d\zeta = \frac{1}{2\pi i}\oint_{|\zeta|=1} \frac{e^{\frac{z}{\zeta}} e^{z\zeta}}{\zeta} d\zeta$

令 $\zeta = e^{i\theta}$, 则 $\sum_{n=0}^{\infty}\left(\frac{z^n}{n!}\right)^2 = \frac{1}{2\pi}\int_0^{2\pi} e^{2z\cos\theta} d\theta$

$= \cos\theta + i\sin\theta$

13 (a) $|z-1|>1 \Rightarrow 0<\left|\frac{1}{z-1}\right|<1$



$$\text{取} \frac{1}{1+\frac{1}{z-1}} = \sum_{n=0}^{\infty}(-1)^n \frac{1}{(z-1)^n}$$

$$\Rightarrow \frac{1}{z} = \sum_{n=0}^{\infty}(-1)^n (z-1)^{-n-1}$$

$$\Rightarrow \frac{1}{z^2} = \sum_{n=0}^{\infty}(-1)^n(-n-1)(z-1)^{-n-2}$$

$$\frac{1}{z^2} = \sum_{n=0}^{\infty}(-1)^n(n+1)(z-1)^{-n-2}$$

$$\text{取} \frac{z-1}{z^2} = \sum_{n=0}^{\infty}(-1)^n(n+1)(z-1)^{-n-1} \qquad (|z-1|>1)$$

(b) $$\frac{1}{z} = \frac{1}{z+2-2} = \frac{1}{2}\cdot\frac{1}{1-\frac{z+2}{2}} = -\frac{1}{2}\sum_{n=0}^{\infty}\left(\frac{z+2}{2}\right)^n = \sum_{n=0}^{\infty} -\frac{(z+2)^n}{2^{n+1}}$$

$$\frac{1}{z(z+2)^3} = \sum_{n=0}^{\infty} -\frac{(z+2)^{n-3}}{2^{n+1}} \qquad (0<|z+2|<2)$$

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